

Weekly Progress Report

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Domain: **Data Science and Machine Learning**

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Week Ending: 01

I. Overview:

This week, the primary focus was on understanding USC_TIA and contributing to Python projects. Additionally, efforts were made to leverage learning resources for skill enhancement.

II. Achievements:

1. USC_TIA Familiarization:

- Explored USC_TIA documentation to grasp core functionalities.
- Successfully executed basic tasks, showcasing initial proficiency.

2. Python Project Contributions:

Name of the project :- Quality Prediction in a Mining Process

- Contributed code to **Quality Prediction in a Mining Process** with a focus on **implementing performance evaluation metrics and testing model accuracy against baseline datasets.**
- Engaged in effective collaboration with team members.

3. Learning Python:

- Acquired proficiency in essential Python libraries, such as Pandas, Matplotlib, Seaborn.
- Applied Python skills to real-world problems within USC_TIA context.

III. Challenges:

1. USC_TIA Integration:

- Encountered challenges during USC_TIA integration with the **historical data repository containing the baseline evaluation datasets**.
- Ongoing efforts to troubleshoot data schema mismatches and ensure successful integration for model validation pipelines.

2. Python Project Complexity:

- Faced complexity in understanding **the alignment of time-series evaluation metrics (such as MAE , RMSE) when comparing model predictions against highly skewed baseline process data** of the Python project.
- Seeking guidance to implement robust cross-validation strategies that avoid data leakage in industrial time-series benchmarks and enhance understanding.

IV. Learning Resources:

1. USC_TIA Documentation:

- Utilized USC_TIA official documentation for reference and troubleshooting.
- Attended relevant webinars and online tutorials to deepen understanding.

2. Python Learning Resources:

- Engaged with **skicit-learn an pandas time-series forecasting tutorials** to strengthen Python skills.
- Participated in **Kaggle industrial processing competitions and Stack Overflow data engineering forums** for practical application.

V. Next Week's Goals:

1. USC_TIA Enhancement:

- Address integration challenges and explore advanced USC_TIA features.
- Collaborate with peers to contribute to USC_TIA improvement discussions.

2. Python Project Development:

- Construct the core evaluation pipeline to calculate **RMSE** , **MAE** against historical mining baseline datasets.
- Implement a chronological **TimeSeriesSplit** loop to test model prediction stability across changing ore feed characteristics.
- Tackle more complex tasks within the Python project to increase contribution.
- Seek feedback from mentors and peers for continuous improvement.

VI. Additional Comments:

- The baseline dataset contains significant skewness due to plant maintenance intervals and sensor downtime records.
- Preliminary evaluations show that slurry intensity and chemical reagent dosing rates have the highest correlation with final quality variations.